

CANDY DIALOGUE ENGINE

Operators

1.1.0

Basic

"="

Assign value directly to the variable.

"+="

Add value to current.

"_="

Subtract value from current.

"*=", "x="

Multiply current by value.

"/=", "÷="

Divide current by value. If value is 0, does nothing (logs an error).

"//="

Integer (floor) division of current by value. If value is 0, does nothing (logs an error).

"%="

Modulo current by value. If value is 0, does nothing (logs an error).

"^=", "=", "pow"**

Raise current to the power of value.

String Operations

"strip_edges"

Has value: strip leading/trailing whitespace from value, assign to variable.

No value: strip leading/trailing whitespace from current.

"to_upper"

Has value: convert value to uppercase, assign to variable.

No value: convert current to uppercase.

"to_lower"

Has value: convert value to lowercase, assign to variable.

No value: convert current to lowercase.

"replace"

Value must be a 2-element array [search, replacement]. Replace all instances of search with replacement in current (returns current unchanged if value isn't a valid array).

"substr"

Value must be a 2-element array [start, length]. Return the substring of current starting at start for length characters (returns current unchanged if value isn't a valid array).

"split"

If current is a string and value is a string, split current by value (returns a Godot PackedStringArray-like split result). Otherwise returns current unchanged.

"split_to_array"

Same as split, but explicitly converts the result to an Array.

"join"

If current is an array and value is a string, join the array elements using value as the separator.

"path_join"

If current and value are both strings, join them as a file path.

Array Operations

"append"

If current is an array, add value to the end (does not mutate original array — works on a duplicate).

"append_first"

If current is an array, insert value at the beginning.

"insert"

Value must be a 2-element array [index, item]. Insert item into current at index.

"erase"

If current is an array, remove the first occurrence of value from it.

"remove_at"

If current is an array and value is a valid integer index, remove the item at that index.

"clear"

Has value: if value is an array, return an empty array; if value is a dictionary, return an empty dictionary; otherwise return current unchanged.

No value: same logic applied to current instead.

"slice"

Value must be a 2-element array [start, end].

Return the slice of current from start to end.

"reverse"

Has value: if value is an array, return a reversed copy of it.

No value: if current is an array, return a reversed copy of it.

"sort"

Has value: if value is an array, return a sorted copy of it.

No value: if current is an array, return a sorted copy of it.

"shuffle"

Has value: if value is an array, return a shuffled copy of it.

No value: if current is an array, return a shuffled copy of it.

"unique"

Has value: if value is an array, return a copy with duplicate elements removed.

No value: same, applied to current.

Dictionary Opeations

"merge"

If current and value are both dictionaries, merge value's keys into a copy of current (value's keys overwrite current's on conflict).

"get"

Value must be either a string (key to look up in current) or a 2-element array [key, default_value].

Returns the corresponding value from the current dictionary, or the default if given and key is missing. Only works if current is a dictionary.

"set_key"

Value must be a 2-element array [key, val]. Set key to val in a copy of current (current must be a dictionary).

"duplicate"

Returns a shallow copy of value if value is an array or dictionary; otherwise returns value unchanged.

"deep_duplicate"

Returns a deep copy of value if value is an array or dictionary; otherwise returns value unchanged.

Type & Conversion

"typeof"

Has value: return the Godot type ID (integer) of value.

No value: return the type ID of current.

"type_string"

Has value: return the human-readable type name of value.

No value: return the human-readable type name of current.

"is_array"

Has value: return true if value is an array.

No value: return true if current is an array.

"is_dict"

Has value: return true if value is a dictionary.

No value: return true if current is a dictionary.

"is_object"

Has value: return true if value is an Object.

No value: return true if current is an Object.

"is_nan"

Has value: return true if value is NaN (as a float).

No value: return true if current is NaN.

"is_inf"

Has value: return true if value is infinite (as a float).

No value: return true if current is infinite.

"to_int"

Has value: convert value to an integer.

No value: convert current to an integer.

"to_float"

Has value: convert value to a float.

No value: convert current to a float.

"to_string"

Has value: convert value to a string.

No value: convert current to a string.

"to_bool"

Has value: convert value to a boolean.

No value: convert current to a boolean.

"toggle_bool"

Has value: return the boolean negation of value.

No value: return the boolean negation of current.

"var_to_str"

Has value: serialize value to its GDScript string representation.

No value: serialize current.

"str_to_var"

Has value: parse value (converted to string) back into a GDScript variant.

No value: parse current the same way.

"json_to_str"

Has value: convert value to a JSON string.

No value: convert current to a JSON string.

"str_to_json"

Has value: parse value (converted to string) as JSON.

No value: parse current as JSON.

"var_to_bytes"

Has value: serialize value to a PackedByteArray.

No value: serialize current.

"bytes_to_var"

If value is a PackedByteArray, deserialize it back into a variant. Otherwise does nothing (logs an error) and returns current.

Math Functions

"abs"

Has value: absolute value of value.

No value: absolute value of current.

"floor"

Has value: floor of value.

No value: floor of current.

"ceil"

Has value: ceiling of value.

No value: ceiling of current.

"round"

Has value: round value to nearest integer.

No value: round current to nearest integer.

"sqrt", "root"

Has value: square root of value. If negative, does nothing (logs an error).

No value: square root of current. If negative, does nothing (logs an error).

"sign"

Has value: sign of value (-1, 0, or 1).

No value: sign of current.

"sin"

Has value: sine of value.

No value: sine of current.

"cos"

Has value: cosine of value.

No value: cosine of current.

"tan"

Has value: tangent of value.

No value: tangent of current.

"asin"

Has value: arcsine of value.

No value: arcsine of current.

"acos"

Has value: arccosine of value.

No value: arccosine of current.

"atan"

Has value: arctangent of value.

No value: arctangent of current.

"deg_to_rad"

Has value: convert value from degrees to radians.

No value: convert current from degrees to radians.

"rad_to_deg"

Has value: convert value from radians to degrees.

No value: convert current from radians to degrees.

"log"

Has value: natural logarithm of value. If value ≤ 0 , does nothing (logs an error).

No value: natural logarithm of current. If current ≤ 0 , does nothing (logs an error).

"log10"

Has value: base-10 logarithm of value. If value ≤ 0 , does nothing (logs an error).

No value: base-10 logarithm of current. If current ≤ 0 , does nothing (logs an error).

"log_base"

If value is a number and not equal to 1, return the logarithm of current in base value. If current ≤ 0 , does nothing (logs an error).

"clamp"

Value must be a 2-element array [min, max]. Clamp current between min and max.

"min"

If value is a non-empty array, return the minimum element of that array. Otherwise return the smaller of current and value.

"max"

If value is a non-empty array, return the maximum element of that array. Otherwise return the larger of current and value.

"lerp"

Value must be a 2-element array [to, weight]. Linearly interpolate from current to "to" by "weight".

"wrap"

Value must be a 2-element array [min, max]. Wrap current between min and max.

Random

"randi", "rand_i"

Has value (as a 2-element array): treated as a range — returns a random integer in that range (deprecated path, warns to use "randi_range" instead).

Has value (not an array): returns current unchanged. No value: return a random integer (full int range).

"randi_range", "rand_i_range"

Has value: if it's a 2-element array, return a random integer in that range.

No value: if current is a 2-element array, return a random integer in that range.

"randf", "rand_f"

Has value (as a 2-element array): treated as a range — returns a random float in that range (deprecated path, warns to use "randf_range" instead).

Has value (not an array): returns current unchanged.

No value: return a random float between 0 and 1.

"randf_range", "rand_f_range"

Has value: if it's a 2-element array, return a random float in that range.

No value: if current is a 2-element array, return a random float in that range.

"rand_pick"

Has value: if value is a non-empty array, return a random element from it.

No value: if current is a non-empty array, return a random element from it.

Nodes

"get_node"

If value is a string, return the node at that path (relative to this node).

"get_node_or_null"

If value is a string, return the node at that path, or null if it doesn't exist.